「Uniqueness

「Thm.

Let ϕ be a real-valued function on $R = [a,b] \times [\alpha,\beta] \subseteq \mathbb{R}^2$.
Consider the initial-value problem

$$\begin{cases} y' = \phi(x,y) \\ y(a) = c \in [\alpha,\beta]. \end{cases}$$

A differentiable function f on $[a,b]$ such that $f(a) = c$ and $\alpha \leq f(x) \leq \beta$ and

$$f'(x) = \phi(x, f(x)) \quad a \leq x \leq b$$

is called a solution for the given problem.
If $y \mapsto \phi(x,y)$ satisfies Lipschitz condition for any $a \leq x \leq b$,
then the given problem has at most one solution.

「Existence.

「Thm.

Let ϕ be a real-valued continuous bounded function on $[0,1]\times\mathbb{R}$.
Then, the initial-value problem

$$\begin{cases} y' = \phi(x,y) \\ y(0) = c \end{cases}$$

has a solution.